
Python for Data Science

Learn Python programming and apply it to real-world data analysis and visualization tasks.

Level: Intermediate | Estimated time: 10 hours | Category: Data

Course overview

Learn Python programming and apply it to real-world data analysis and visualization tasks.

This PDF course pack was generated for the project from publicly available internet resources and the app's built-in lesson outline.

Lesson 1: Python Fundamentals

Python is a high-level, interpreted programming language known for its clear syntax.

- Installing Python and setting up a virtual environment
- Variables and data types: int, float, str, bool, list, dict, tuple, set
- Operators and expressions
- String formatting: f-strings, .format()
- Conditional statements: if, elif, else
- Loops: for, while, list comprehensions
- Functions: def, parameters, return, *args, **kwargs
- Modules and imports
- File I/O: open(), read(), write()
- Exception handling: try/except/finally

Key concept: Python uses indentation (whitespace) to define code blocks - there are no curly braces.

Lesson 2: NumPy & Pandas

NumPy and Pandas are the core libraries for data science in Python.

NumPy: - Arrays: np.array(), shape, dtype - Array operations: slicing, indexing, broadcasting - Math functions: np.sum(), np.mean(), np.std()

Pandas: - Series and DataFrame - Reading data: pd.read_csv(), pd.read_excel() - Exploring data: .head(), .info(), .describe() - Selecting data: loc, iloc, boolean indexing - Handling missing data: .isnull(), .dropna(), .fillna() - Grouping and aggregating: .groupby(), .agg()

Key concept: A DataFrame is like a spreadsheet in Python - rows are observations, columns are variables.

Further reading from the internet

These official or widely used references match the course topic and can be used for deeper study.

Python tutorial

<https://docs.python.org/3/tutorial/>

NumPy quickstart

<https://numpy.org/doc/stable/user/quickstart.html>

pandas getting started

https://pandas.pydata.org/docs/dev/getting_started/index.html